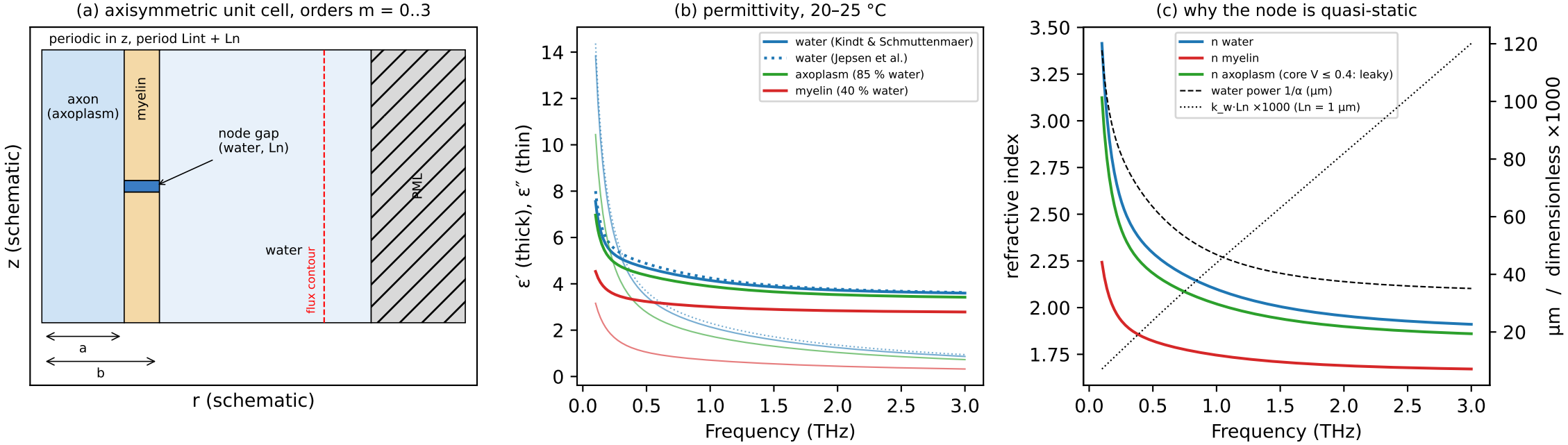
(a) axisymmetric unit cell, orders m = 0..3
periodic in z, period Lint + Ln
axon
(axoplasm)
myelin
node gap
(water, Ln)
water
flux contour
RML
a
b
z (schematic)
r (schematic)
(b) permittivity, 20–25 °C
water (Kindt & Schmuttenmaer)
water (Jepsen et al.)
axoplasm (85 % water)
myelin (40 % water)
ε′ (thick), ε″ (thin)
Frequency (THz)
(c) why the node is quasi-static
n water
n myelin
n axoplasm (core V ≤ 0.4: leaky)
water power 1/α (μm)
k_w·Ln ×1000 (Ln = 1 μm)
refractive index
μm / dimensionless ×1000
Frequency (THz)